International Baccalaureate®
Baccalauréat International
Bachillerato Internacional

BIOLOGY
HIGHER LEVEL
PAPER 2

Candidate session number

SPECIMEN PAPER

2 hours 15 minutes

Examination code

				–				

INSTRUCTIONS TO CANDIDATES

- Write your session number in the boxes above.
- Do not open this examination paper until instructed to do so.
- Section A: answer all questions.
- Section B: answer two questions.
- Write your answers in the boxes provided.
- A calculator is required for this paper.
- The maximum mark for this examination paper is *[72 marks]*.

20 pages

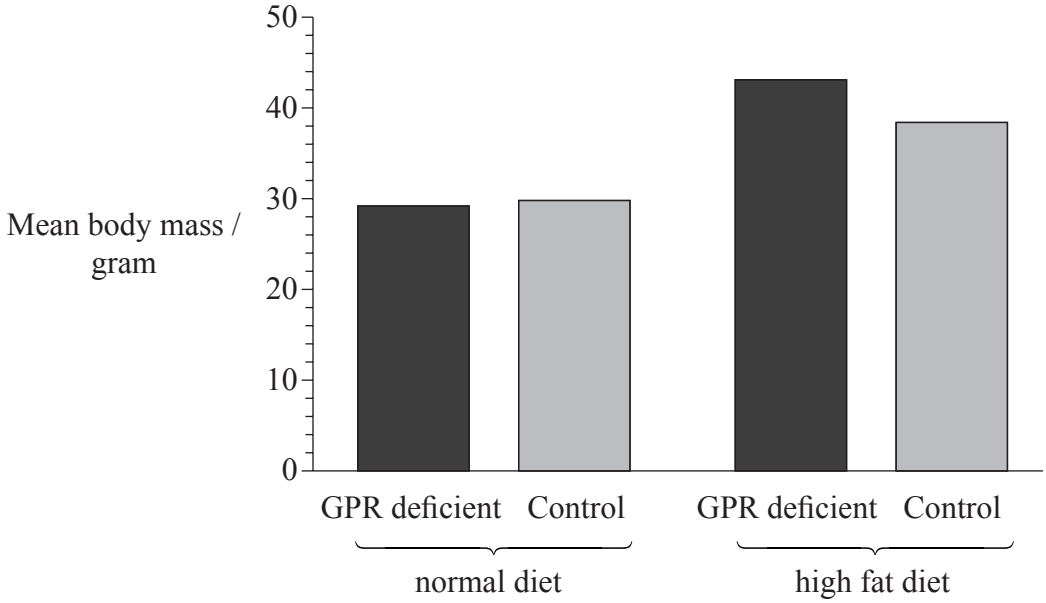
© International Baccalaureate Organization 2014

SECTION A

*Answer **all** questions. Write your answers in the boxes provided.*

1. GPR120 is a trans-membrane protein that functions as a receptor for long-chain unsaturated fatty acids. When fatty acids outside the cell bind to the receptor, the receptor changes shape and as a result a signal is passed to the interior of the cell. This causes an increase in the intracellular calcium concentration (Ca^{2+}), which has a wide range of effects on cell activity.

A strain of mice was developed that did not produce GPR120 protein. Groups of these GPR120 deficient mice were fed from the age of 5 weeks until they were 16 weeks old either on a high fat diet containing 60 % fat or on a normal diet containing 13 % fat. Control groups of mice that did produce GPR120 protein were fed on the same diets. The bar chart shows the mean body mass of each group when the mice were 16 weeks old.



[Source: adapted from A Ichimur, *et al.*, (2012), *Nature*, **483**, pages 350–354, Reprinted by permission from Macmillan Publishers Ltd]

(This question continues on the following page)

(Question 1 continued)

(a) Compare and contrast the body mass of GPR120 deficient mice and the control mice on a normal diet and a high fat diet. *[3]*

. . . . . .

. . . . . .

. . . . . .

. . . . . .

. . . . . .

. . . . . .

(This question continues on the following page)

(Question 1 continued)

The gene for GPR120 protein is located on Chromosome 10. The alleles for GPR120 protein from 312 extremely obese adults and children were base sequenced. Six alleles were discovered that differed by one base from the wild-type allele. They are likely to have been produced by a base substitution mutation. Two of the mutations cause a change in the amino acid sequence of the GPR120 protein (missense mutation) but the other four do not (synonymous mutation). The table gives details of the six mutant alleles.

Mutant allele	**Nucleotide change**	**Chromosome 10 position**	**Type of mutation**
R67C	C → T	95 316 666	Missense
R270H	G → A	95 337 031	Missense
V38V	G → A	95 316 581	Synonymous
S192S	G → A	95 325 846	Synonymous
V243V	C → T	95 328 938	Synonymous
S264S	G → A	95 337 014	Synonymous

[Source: adapted from A Ichimur, *et al.*, (2012), *Nature*, **483**, pages 350–354, Reprinted by permission from Macmillan Publishers Ltd]

(b) Explain how only some of the base substitution mutations cause a change in the amino acid sequence of the gene for GPR120. *[2]*

. . . . . .

. . . . . .

. . . . . .

. . . . . .

(c) Using the positions on chromosome 10, identify which **two** mutations were closest together in the gene for GPR120. *[1]*

1. . . . . .

2. . . . . .

(This question continues on the following page)

(Question 1 continued)

The frequency of the two alleles that caused a change in amino acid sequence was measured in 6942 unrelated obese humans and 7654 controls. The results are shown in the table.

Allele	Allele frequency / %	
	obese humans	controls
R67C	5.5	4.3
R270H	2.4	1.3

[Source: adapted from A Ichimur, *et al.*, (2012), *Nature*, **483**, pages 350–354, Reprinted by permission from Macmillan Publishers Ltd]

(d) Outline the reasons for using large numbers of obese humans and controls in this research. *[2]*

. . . . . .

. . . . . .

. . . . . .

. . . . . .

(e) The association of one of the alleles with obesity was statistically significant. Deduce, with a reason, which allele this was. *[1]*

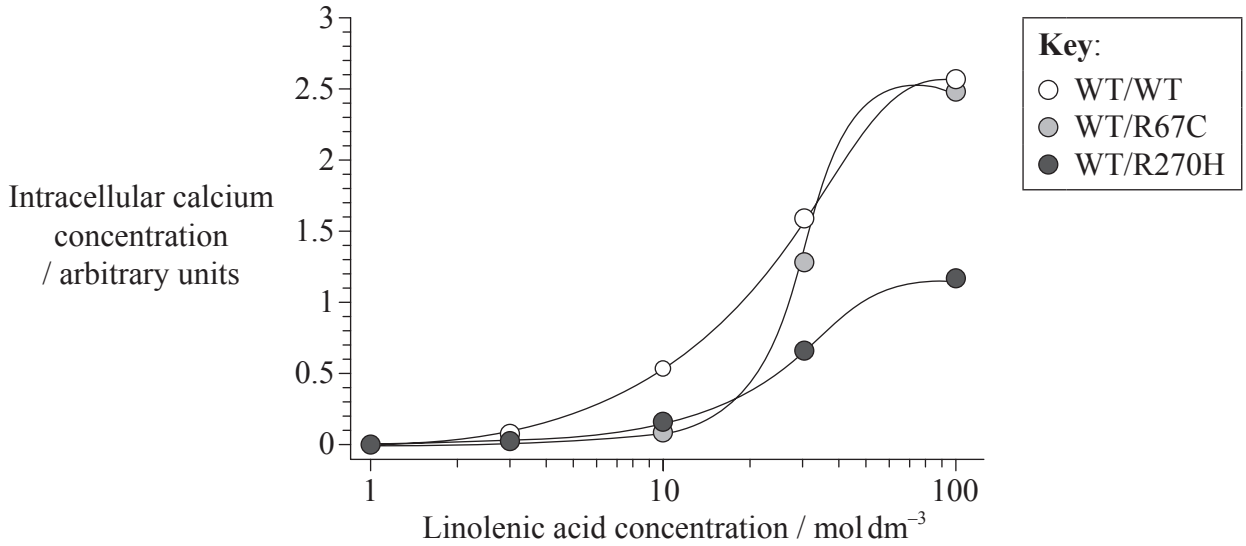
. . . . . .

. . . . . .

(This question continues on the following page)

(Question 1 continued)

A clone of human cells that contained the wild type allele for GPR120 was genetically modified by inserting either another wild-type allele (WT) or one of two mutant alleles (R67C and R270H). The intracellular concentration of calcium was measured in these cells at varying levels of linolenic acid. Linolenic acid is a long-chain unsaturated fatty acid. The graph shows the results. The scale on the *x*-axis is logarithmic.



(f) Suggest advantages of genetically modifying a clone of human cells for use in this experiment rather than using cells from obese people naturally containing the mutant alleles. *[2]*

. . . . . .

. . . . . .

. . . . . .

. . . . . .

(This question continues on the following page)

(Question 1 continued)

(g) Outline the effect of linolenic acid concentration on intracellular calcium concentration in the WT/WT cells. *[2]*

. . . . . .

. . . . . .

. . . . . .

. . . . . .

(h) Compare and contrast the effect of the two mutant alleles on intracellular calcium concentration. *[2]*

. . . . . .

. . . . . .

. . . . . .

. . . . . .

Biologists have often debated whether characteristics in humans are due to genes, the environment, or a combination of both.

(i) Discuss the evidence provided by the data for the relative role of genes and diet in causing obesity. *[3]*

. . . . . .

. . . . . .

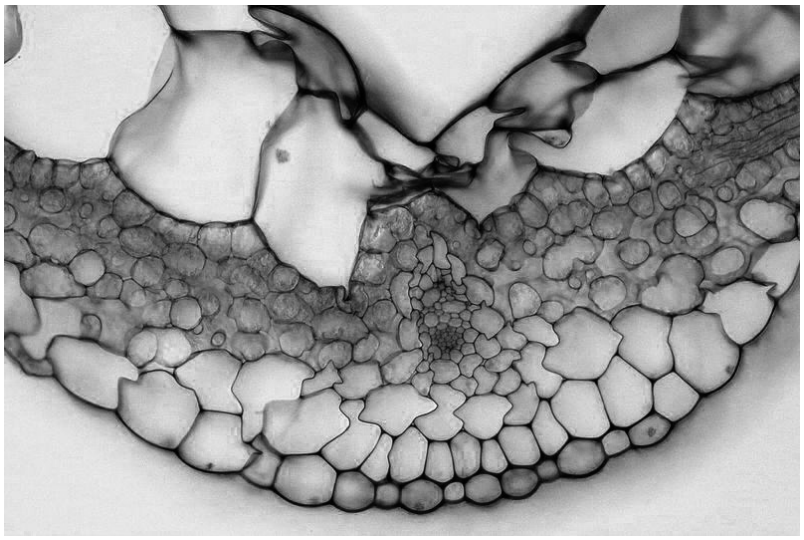
. . . . . .

. . . . . .

. . . . . .

. . . . . .

2. The micrograph below shows a thin vertical section through a *Tradescantia* leaf.



[Source: adapted from http://upload.wikimedia.org/wikipedia/commons/thumb/5/5b/Tradescantia%2C_leaf%2C_Etzold_green_1.JPG/800px-Tradescantia%2C_leaf%2C_Etzold_green_1.JPG]

(a) State **two** general features of the structure of complex organisms that are visible in the micrograph. *[2]*

1. . . . . .

2. . . . . .

The leaf tissue contains the enzyme Rubisco and NADP.

(b) Outline the function of Rubisco. *[2]*

. . . . . .

. . . . . .

. . . . . .

. . . . . .

(This question continues on the following page)

(Question 2 continued)

(c) Explain precisely where NADP is located in the leaf and how it is used. *[4]*

Location:

. . . . . .

. . . . . .

. . . . . .

Function:

. . . . . .

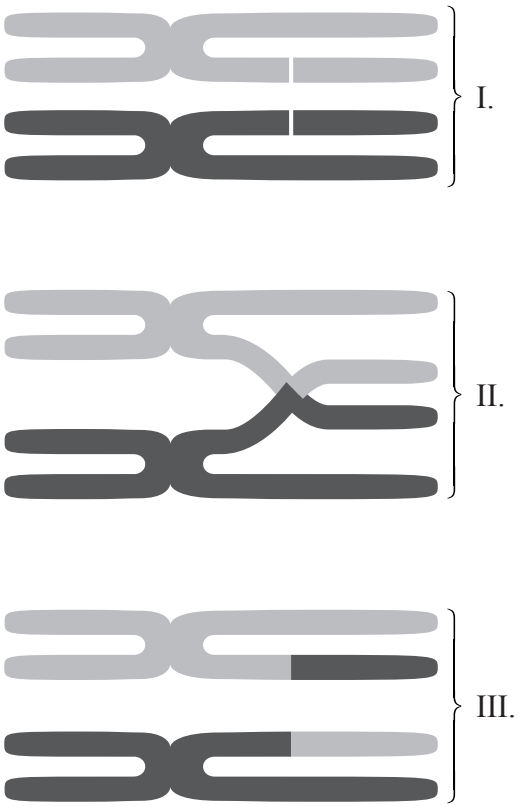
. . . . . .

. . . . . .

. . . . . .

. . . . . .

3. The diagrams show two chromosomes at three stages in meiosis.



(a) The two chromosomes in these diagrams are homologous. State **one** similarity and **one** difference between homologous chromosomes. *[2]*

Similarity:

. . . . . .

. . . . . .

Difference:

. . . . . .

. . . . . .

(This question continues on the following page)

(Question 3 continued)

(b) State, with a reason, whether the cell that contained these chromosomes would have been haploid **or** diploid. *[1]*

. . . . . .

. . . . . .

(c) Explain how genetic variation is promoted by the process shown in the diagrams. *[3]*

. . . . . .

. . . . . .

(d) State the type of life-cycle that includes meiosis and the reason for it being needed in this type of life-cycle. *[2]*

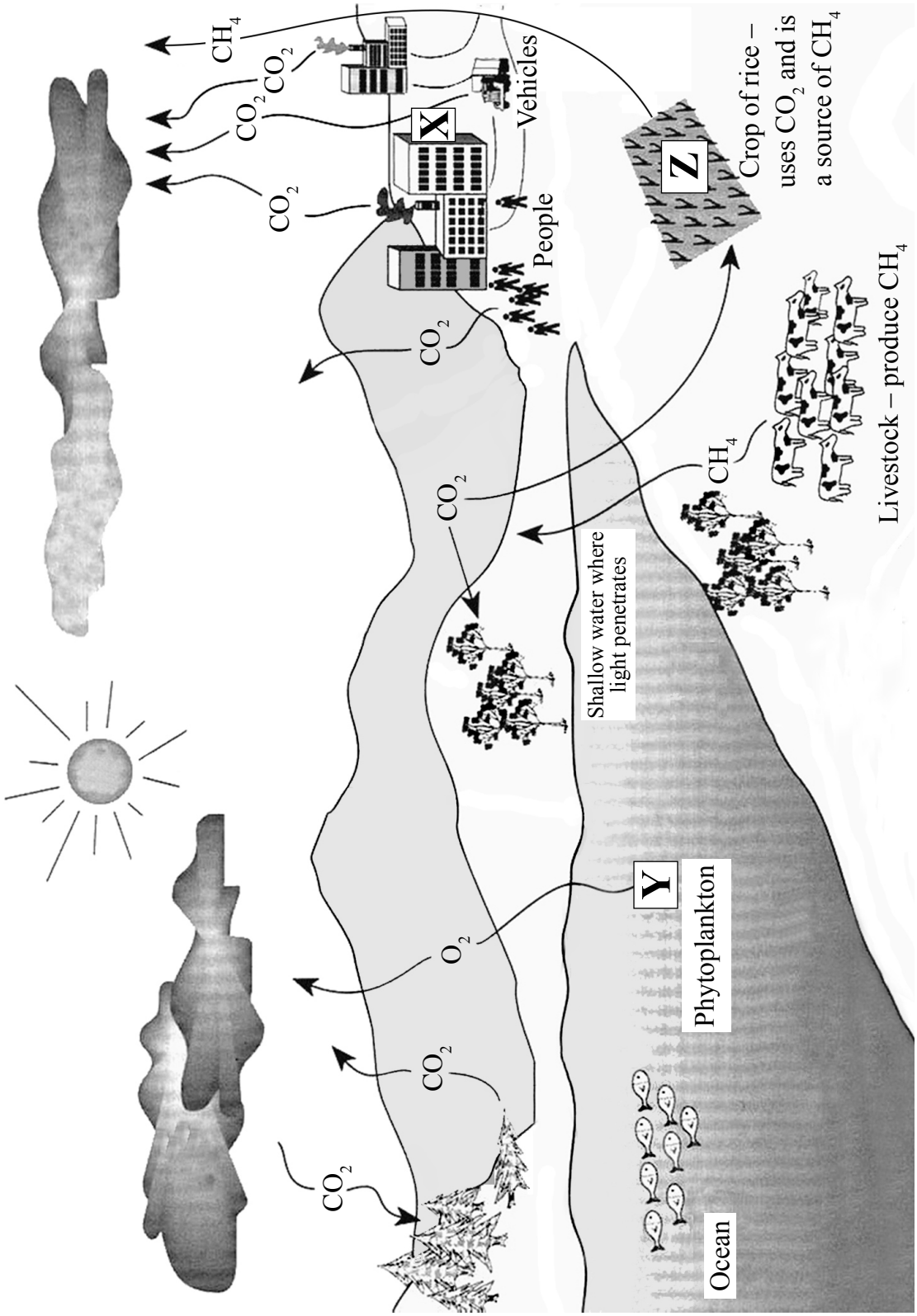
. . . . . .

. . . . . .

. . . . . .

. . . . . .

4. The diagram shows the carbon cycle.



[Source: adapted from www-das.uwyo.edu/~geerts/cwx/notes/chap01/carbon_cycle.jpeg]

(This question continues on the following page)

(Question 4 continued)

(a) State the processes occurring at X and Y. *[2]*

X: . . . . . .

Y: . . . . . .

(b) Predict the conditions that would increase the release of methane shown at Z. *[2]*

. . . . . .

. . . . . .

. . . . . .

. . . . . .

(c) Outline the impact of the gases shown in the diagram on the greenhouse effect. *[2]*

. . . . . .

. . . . . .

. . . . . .

. . . . . .

SECTION B

*Answer **two** questions. Up to [1] additional mark is available for the quality of your answers for each question.*

5. Water has properties that make it essential to all living organisms.

(a) Explain how the properties of water, that are essential to living things, arise from the dipolar nature of water. *[8]*

(b) Describe the adaptations of plants in deserts for water conservation. *[4]*

(c) Outline the use of models to investigate the transport of water in xylem. *[3]*

6. From an early stage in human development, blood flow is needed for transport and exchange of materials.

(a) Explain the exchange of materials between the mother and the fetus in the uterus. *[8]*

(b) Outline how William Harvey changed the understanding of blood flow around the human body. *[4]*

(c) Describe how, in microscope images, blood vessels can be identified as arteries, capillaries and veins. *[3]*

7. The species is the basis for naming and classifying organisms.

(a) Explain how new species can emerge by

- directional selection
- disruptive selection
- polyploidy. *[8]*

(b) Outline the advantages to scientists of the binomial system for naming species. *[4]*

(c) Describe the use of dichotomous keys for the identification of specimens. *[3]*